ST5202 — Applied Regression Analysis
Lecture Notes 5. Other Special multiple LRMs

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Outline

§1. Analysis of covariance (ANOCOV) models

§2. LRM with non-linear predictor terms

§1.1. LRM with both factor and quantitative predictors

- ANOCOV is a historical terminology. It arises from the analysis of the effects of factors while there are certain quantitative variables which also have effect on the response variable.
- An ANOCOV Model is indeed a special multiple LRM that has both factor and quantitative predictors, e.g.,

$$y_i = \beta_0 + \sum_{j=2}^{k} \alpha_j u_{ij} + \gamma x_i + \epsilon_i,$$

  where $u_j, j = 2, \ldots, k$, are dummy variables representing a factor predictor and x is a quantitative predictor.
- In general, an ANOCOV model could have more than one factors and more than one quantitative predictors. It could contain the interaction terms among factors, among quantitative predictors, or between factors and quantitative predictors.

§1.2. Applications of ANOCOV models

- ANOCOV models are mainly used for two practical situations:
  - (i) In the comparison of treatment effects of factor predictors, the effect of certain concomitant variables, which have effect on the response as well, need to be adjusted so that the comparison is more reliable and efficient.
  - (ii) In certain problems, one is primarily concerned with the relationship between the response variable and quantitative predictors, but there are different categories under which the relationship is to be studied. The regression relationship is to be compared among the categories.
- Traditionally, the terminology ANOCOV is narrowly used for case (i). In case (ii), it is called the comparison of regression functions.

§1.3. A remark on the traditional approach to ANOCOV

- The traditional approach to ANOCOV (see, e.g., Montgomery, Design and Analysis of Experiments) is, like ANOVA, based on the decomposition of SST into the SS due to different effects of the factors, but the SS are adjusted for the effect of the concomitant variables. The adjusted SS have complicated formulae involving sample variances and covariances.
- The traditional approach to ANOCOV has the same limitations as that to ANOVA.
- By using the regression approach, when estimating the factor effects, the effect of the concomitant variables is automatically adjusted. It avoids the drawbacks of the traditional approach.

§1.4. Cracker promotion (CP) example

- A company studied the effects of three different types of promotions on sales of its crackers. Fifteen stores were selected for the study. Each store was randomly assigned one of the promotion types, with five stores for each promotion type. Data consists of the number of cases of the product sold in the promotion period (Y) and the number of cases of the product sold in the preceding period (X).
- The data is provided in the file `CrackerSale.txt` on Canvas.
- The main interest is on the effects of the promotion types. X is a concomitant variable, which has a potential effect on Y.

§1.5. CP example: Model and interpretation

- The model:

$$y_i = \beta_0 + \sum_{j=2}^{3} \alpha_j u_{ij} + \gamma x_i + \epsilon_i,$$

  where u_{i2} and u_{i3} are dummy variables for type 2 and type 3, type 1 is taken as the reference level.
- Interpretation of the parameters: α_j, $j = 2, 3$, is the difference of sales between type j and type 1 when X is the same for both types. γ is the effect of X on Y, when X changes by a unit, Y has an expected change γ.

§1.6. CP example: R codes

- R codes:

```
CS.dat=read.table("D://Rsession/CrackerSale.txt",header=TRUE)
CS.dat$type=factor(CS.dat$type)
CS.fit = lm(y~x+type,data=CS.dat)
summary(CS.fit); anova(CS.fit)
```

- Remark on the R codes: The ANOVA table produced by `anova` depends on the order of the variables in the formula specification. Only the second factor (or variable) has the valid SS, F and p-values. It is recommended to put the concomitant variable first followed by the factor.
- The correct F value can always be obtained through Wald statistics regardless of the order of the variables.

§1.7. CP example: Fitted results

- Results with variable order: `x, type`.

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  17.3534     2.5230   6.878 2.66e-05
x             0.8986     0.1026   8.759 2.73e-06
type2        -5.0754     1.2290  -4.130  0.00167
type3       -12.9768     1.2056 -10.764 3.53e-07

          Df Sum Sq Mean Sq F value    Pr(>F)
x          1 190.68 190.678  54.379 1.405e-05
type       2 417.15 208.575  59.483 1.264e-06
Residuals 11  38.57   3.506
```

- Results with variable order: `type, x`.

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  17.3534     2.5230   6.878 2.66e-05
type2        -5.0754     1.2290  -4.130  0.00167
type3       -12.9768     1.2056 -10.764 3.53e-07
x             0.8986     0.1026   8.759 2.73e-06

          Df Sum Sq Mean Sq F value    Pr(>F)
type       2 338.80 169.400  48.310 3.566e-06
x          1 269.03 269.029  76.723 2.731e-06
Residuals 11  38.57   3.506
```

- F-value for type computed by:

```
b=CS.fit$coef[c(2,3)];V=vcov(CS.fit)[c(2,3),c(2,3)]
F=t(b)%*%solve(V)%*%b/2
```

$F = 59.48282$.

§1.8. CP example: conclusion

- The promotion type has significant effects on the number of sales, since the F-statistic for type has a p-value `1.264e-06` from the ANOVA table.
- Both type 2 and type 3 are significantly different from type 1, judging by the p-values 0.00167 and `3.53e-07` from the summary table. Specifically, type 2 is worse than type 1, and type 3 is worse than type 2.
- The sales in the preceding period has a significant effect on the promotion sales with a p-value `2.73e-06` from the summary table.

§1.9. CP exmaple: what if without adjustment?

- Without adjusting for the effect of X, the fitted results are:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   38.200      2.264  16.871 1.01e-09
type2         -2.200      3.202  -0.687  0.50511
type3        -11.000      3.202  -3.435  0.00494

          Df Sum Sq Mean Sq F value  Pr(>F)
type       2  338.8 169.400  6.6086 0.01161
Residuals 12  307.6  25.633
```

- We can still conclude that the promotion type has significant effect. But the significance of the difference between type 2 and type 1 cannot be concluded.
- Without adjusting the effect of X, the results could be misleading.

§1.10. Salable flowers (SF) example

- A horticulturist conducted an experiment to study the effects of flower variety (factor A: varieties LP, WB) and moisture level (factor B: low, high) on yield of salable flowers (Y). Since the plots were not of the same size, the plot size (X), which might have an effect on Y, was used as a concomitant variable.
- The data is given in the file `SalableFlower.txt` on Canvas.
- To investigate possible interaction effect between flower variety and moisture, an interaction model is used.

§1.11. SF example: Model and interpretation

The expected yields expressed at each combination of the factor levels:

$$EY = \begin{cases} \beta_0 + \gamma X, & \text{LP-Low;} \\ \beta_0 + \alpha_2 + \gamma X, & \text{WB-Low;} \\ \beta_0 + \beta_2 + \gamma X, & \text{LP-High;} \\ \beta_0 + \alpha_2 + \beta_2 + \gamma_{22} + \gamma X, & \text{WB-High.} \end{cases}$$

- α_2: Difference between variety WB and LP with low moisture when plot size is the same.
- $\alpha_2 + \gamma_{22}$: Difference between variety WB and LP with high moisture when plot size is the same.
- β_2: Difference between high and low moisture for variety LP when plot size is the same.
- $\beta_2 + \gamma_{22}$: Difference between high and low moisture for variety WB when plot size is the same.
- γ_{22}: Interaction effect when plot size is the same.

§1.12. SF example: ANOVA tables

```
SF.dat = read.table("D://Rsession/SalableFlower.txt",header=TRUE)
SF.dat$A=factor(SF.dat$A); SF.dat$B=factor(SF.dat$B)
AB.fit=lm(y~x+A*B,data=SF.dat); anova(AB.fit)
BA.fit=lm(y~x+B*A,data=SF.dat); anova(BA.fit)
```

```
          Df Sum Sq Mean Sq  F value    Pr(>F)
x          1 4532.6  4532.6 720.7836 < 2.2e-16
A          1   93.4    93.4  14.8536  0.001069
B          1  324.4   324.4  51.5918 7.991e-07
A:B        1   16.0    16.0   2.5511  0.126719
Residuals 19  119.5     6.3
```

```
          Df Sum Sq Mean Sq  F value    Pr(>F)
x          1 4532.6  4532.6 720.7836 < 2.2e-16
B          1  320.3   320.3  50.9327 8.749e-07
A          1   97.6    97.6  15.5127 0.0008815
B:A        1   16.0    16.0   2.5511 0.1267191
Residuals 19  119.5     6.3
```

§1.13. SF example: inference on significance of main and interaction effects

- To make inference on main effects, the ANOVA table produced by `anova` must be used in caution. Only the factor appeared second has the valid sum of squares no matter whether or not the data is balanced (the current example has a balanced data set). For this reason, the model should be fitted twice with the factors in different orders.
- From the ANOVA tables, it can be concluded that the interaction effect, which has a p-value 0.1267, is not significant. The main effects of A, which has a p-value 0.00088, and of B, which has a p-value `7.99e-07`, are significant.

§1.14. SF example: inference on effect of one factor at fixed level of another factor (1)

- The concomitant effect and the effect of one factor at fixed level of another factor can be inferred using the summary table and the variance matrix of the fitted model (order of the factors does not matter). The table from `AB.fit` is as follows:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   49.508      1.555  31.842  < 2e-16
x              3.277      0.130  25.203 4.59e-16
A2            -5.723      1.454  -3.937 0.000885
B2            -9.000      1.448  -6.216 5.69e-06
A2:B2          3.277      2.052   1.597 0.126719
```

- $\hat{\gamma} = 3.277$, the estimated effect of plot size, is significant (with p-value `4.593-16`).
- $\hat{\alpha}_2 = -5.723$, the estimated difference between variety WB and LP when the moisture level is low, is significant (with p-value 0.000885).
- $\hat{\beta}_2 = -9$, the estimated difference between high and low moisture level when the variety is LP, is significant (with p-value `5.69e-06`)

§1.15. SF example: inference on effect of one factor at fixed level of another factor (2)

- To make inference on other level effects, i.e., the difference between variety WB and LP when the moisture level is high, the variance matrix of $(\hat{\alpha}_2, \hat{\beta}_2, \hat{\gamma}_{22})$ is required. It is extracted from `AB.fit` and given below:

```
                A2        B2     A2:B2
A2      2.113066  1.048081 -2.079257
B2      1.048081  2.096161 -2.096161
A2:B2 -2.079257 -2.096161  4.209227
```

- The difference between variety WB and LP when the moisture level is high: $\alpha_2 + \gamma_{22}$. The test statistic:
  $T = (-5.723 + 3.277)/[2.1131 + 4.2092 + 2 * (-2.0793)]^{1/2} = -1.6629.$
  The p-value: $2P(t_{19} > 1.6629) = 0.112747$. The difference is not significant.
- The similar inference can be made on the difference between high moisture level and low moisture level when the variety is WB, $\beta_2 + \gamma_{22}$.

§1.16. Comparison of regression functions

- When the regression relationship between the response and covariates is studied in different groups (determined by the levels of a factor), it is of interest to compare: (i) whether the regression functions have the same intercepts, (ii) whether they have the same slopes.
- It is not efficient to study them separately in each of the groups. In stead, a single ANOCOV model incorporating the group factor into the regression function is more appropriate.
- Consider the model

$$Y = \beta_0 + \sum_{j=2}^{k} \alpha_j U_j + \beta X + \sum_{j=2}^{k} \xi_j U_j X + \epsilon.$$

Group-wise, the model is expressed as

$$Y = \beta_0 + \beta X + \epsilon, \text{ for category 1,}$$
$$Y = (\beta_0 + \alpha_j) + (\beta + \xi_j) X + \epsilon, \text{ for category } j \ (\geq 2).$$

The problem (i) and (ii) are boiled down to the inference on the α_j's and ξ_j's.

§1.17. 2004 US presidential election (PE) example

- The observations in 67 counties of Florida were obtained on percentage of votes in 2000 and 2004 cast for Bush. In 2004, e-voting was introduced. How the percentage of votes in 2000 affects the change in the percentage of votes from 2000 to 2004 is to be studied in e-voting and non e-voting voters.
- The data is provided in the file `USelection.txt` on Canvas that contains `e.vote` (factor V: 0 for no e-voting, 1 for e-voting), `Bush.pct.2000` (X): percentage of votes for Bush in 2000, `Change.in.Bush.pct` (Y): change in the percentage of votes for Bush.
- The problem is studied by the following model:

$$Y = \beta_0 + \alpha_2 V + \beta X + \xi VX + \epsilon.$$

§1.18. PE example: R codes and results

- R codes:

```
PE.dat=read.table("D://Rsession/USelection.txt",header=TRUE)

y=PE.dat$Change.in.Bush.pct; x=PE.dat$Bush.pct.2000;
V=PE.dat$e.vote;  V=factor(V)

PE.fit = lm(y~x*V); summary(PE.fit)
```

- Fitted results:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.23862    2.35084  -2.228 0.029431 *
x            0.16228    0.04051   4.006 0.000166 ***
V1           9.67236    4.26530   2.268 0.026787 *
x:V1        -0.20051    0.07789  -2.574 0.012403 *
```

§1.19. PE example: Fitted regression lines and comparison

- The fitted regression lines:

$$\begin{aligned} \text{No e-voting}: &\quad EY = \hat{\beta}_0 + \hat{\beta}X = -5.2386 + 0.1623X; \\ \text{e-voting}: &\quad EY = (\hat{\beta}_0 + \hat{\alpha}_2) + (\hat{\beta} + \hat{\xi})X = 4.43374 - 0.03823X. \end{aligned}$$

  The two regresion lines are quite different.
- Since $\hat{\beta}$ is significant with p-value 0.00017, the percentage of votes in 2000 has a significant linear relation with the change of the percentages.
- From the summary table, $\hat{\alpha}_2$ and $\hat{\xi}$ are significant with p-values, respectively, 0.0268 and 0.0124. The difference of the regression lines are significant.
- It can be concluded that e-voting has a negative effect on the votes for Bush. A possible reason behind might be that the republicans have lower intention to go e-voting.

§2.1. What is actually meant by "linear" in LRM?

- A linear regression model is given by
$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon,$$
where the X's are referred to as covariates or predictors.
- A distinction must be made between covariates and physical variables. A physical variable is something like blood pressure, temperature, humility measurement etc. A covariate could be a physical variable or a function of physical variables.
- The restrictive adjective "linear" refers to the linearity of the regression parameters, and the linearity of the covariates, not of the physical variables.
- In a particular case, X_j can be taken as Z^j for a single physical variable Z. The resultant model
$$Y = \beta_0 + \beta_1 Z + \beta_2 Z^2 + \cdots + \beta_p Z^p + \epsilon$$
is a polynomial regression model.
- Let $\boldsymbol{Z}$ be a vector of predictor variables. In general, X_j can be taken as $\phi_j(\boldsymbol{Z})$ where ϕ_j can be any function. The resultant LRM is
$$Y = \beta_0 + \beta_1 \phi_1(\boldsymbol{Z}) + \cdots + \beta_p \phi_p(\boldsymbol{Z}) + \epsilon.$$

§2.2. Polynomial regression models

- A polynomial regression model can involve more than one predictor variables, e.g.,

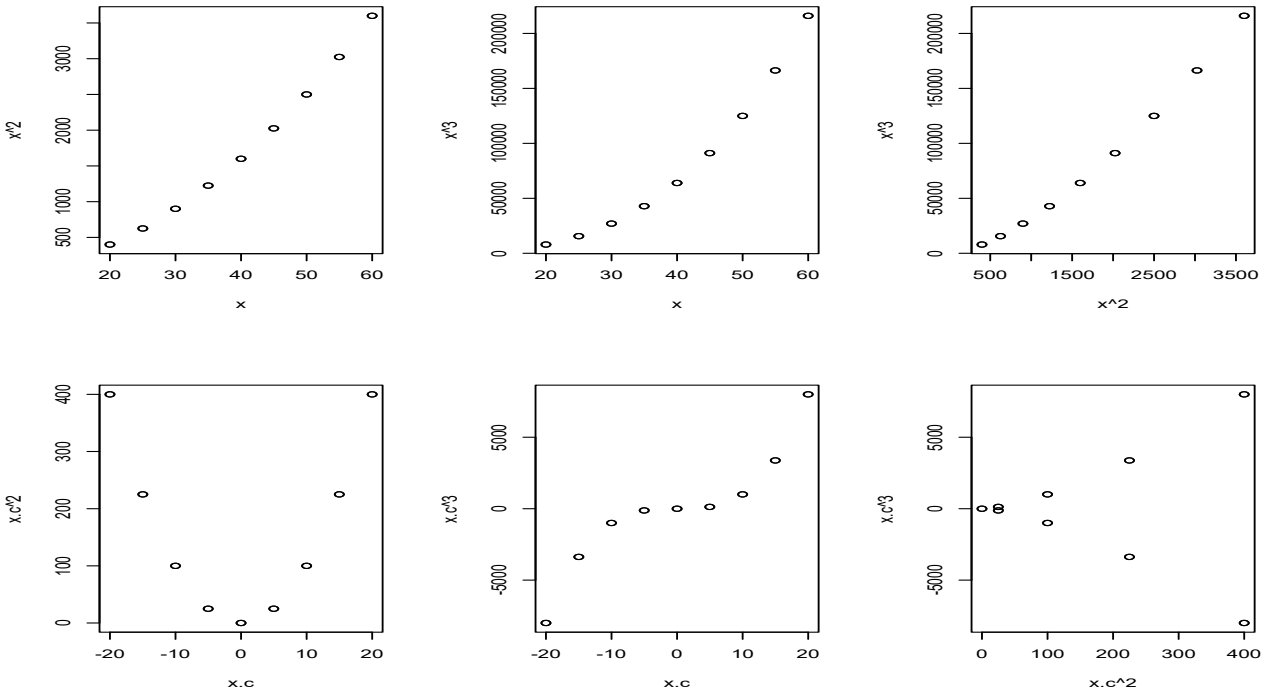
$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i}^2 + \beta_4 x_{2i}^2 + \epsilon_i, \ \ i = 1, \ldots, n.$$

- In dealing with polynomial regression models, it is better to center each of the physical variables, i.e., to consider

$$y_i = \beta_0 + \beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + \beta_3 (x_{1i} - \bar{x}_1)^2 + \beta_4 (x_{2i} - \bar{x}_2)^2 + \epsilon_i.$$

- The centralization is to reduce the multicollinearity of the model, an issue that will be addressed in more detail in later lectures.

§2.3. The effect of centralization to reduce multicollinearity



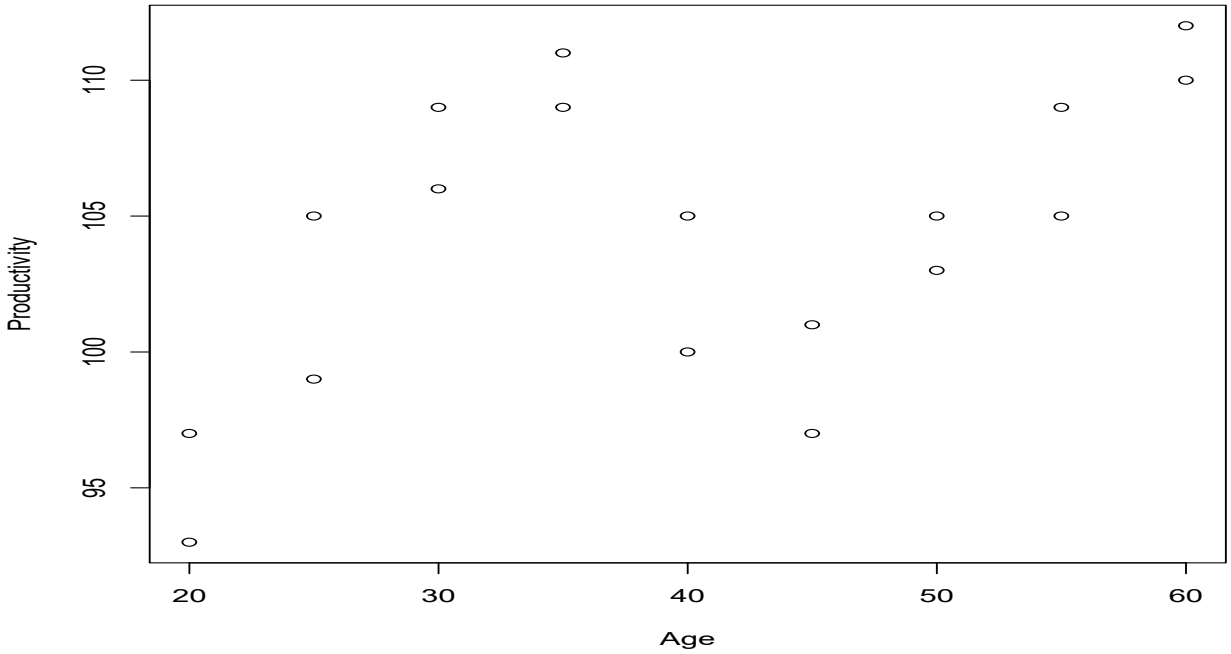
The upper panel shows the correlation for un-centralized linear, quadratic and cubic terms. The lower panel shows the correlation of centralized terms. The correlation is broken up or reduced in the centralized terms.

§2.4. Piecework operation (PO) example

- An operation analyst in a multinational electronics firm studied factors affecting production in a piecework operation where earnings are based on the number of pieces produced. The particular concern is on the effect of the employee's age (X) on their productivity (Y).
- The data is provided in the file `Piecework.txt` on Canvas.
- R codes:

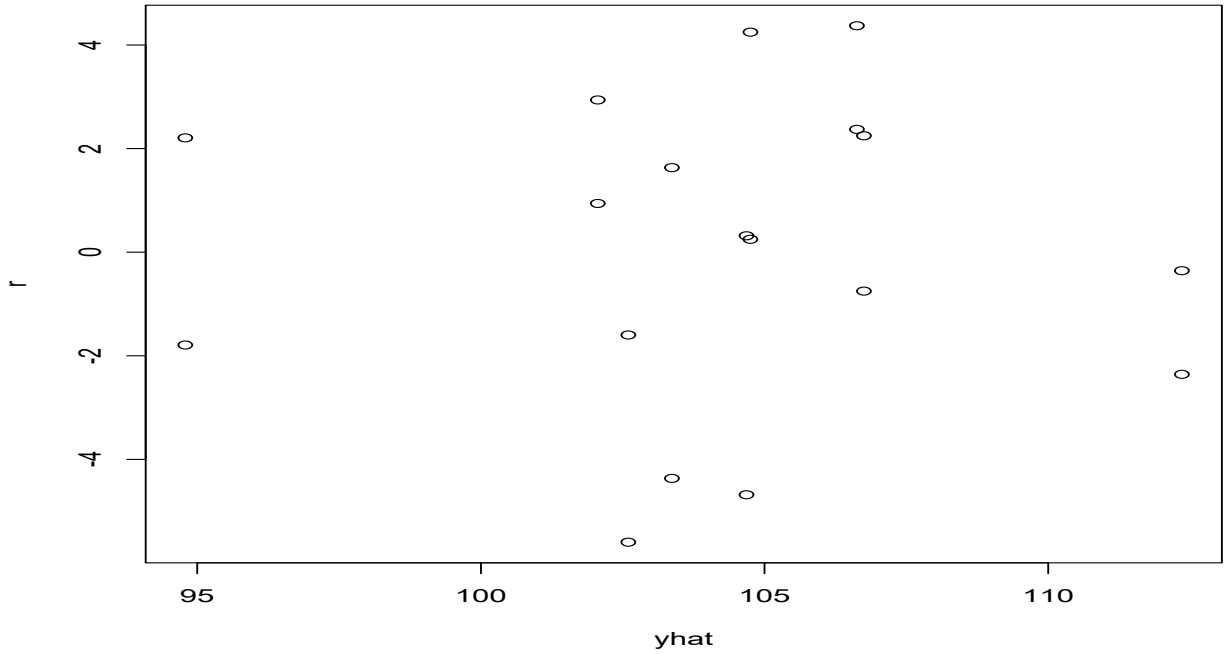
```
PO.dat=read.table("D://Rsession/piecework.txt",header=TRUE)
y=PO.dat$y; x=PO.dat$x; x.c=x-mean(x)
PO.fit = lm(y~x.c+I(x.c^2)+ I(x.c^3),data=PO.dat)
summary(PO.fit)
```

§2.5. PO example: scatter plot



The plot suggests that a cubic regression model might be appropriate.

§2.6. PO example: residual plot



The residual plot does not show any trend and curvature, indicating that the cubic regression model is adequate.

§2.7. PO example: fitted results and conclusion

- Fitted results:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.047e+02  1.168e+00  89.664  < 2e-16
I(x.c)      -4.592e-01  1.562e-01  -2.939 0.010776
I(x.c^2)    -2.771e-03  5.265e-03  -0.526 0.606993
I(x.c^3)     2.246e-03  4.895e-04   4.588 0.000422
Residual standard error: 3.267 on 14 degrees of freedom
Multiple R-squared:  0.7018, Adjusted R-squared: 0.6379
F-statistic: 10.98 on 3 and 14 DF,  p-value: 0.0005662
```

- Conclusion: The productivity and employee's age have a significant cubic relationship.

§2.8. PO example: cubic regression without centralization

- R codes

```
PO.fit.x = lm(y~x+I(x^2)+ I(x^3),data=PO.dat)
summary(PO.fit.x)
```

- Results:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.511e+01  2.676e+01  -0.938 0.363924
x            1.054e+01  2.246e+00   4.694 0.000345
I(x^2)      -2.723e-01  5.897e-02  -4.617 0.000399
I(x^3)       2.246e-03  4.895e-04   4.588 0.000422
Residual standard error: 3.267 on 14 degrees of freedom
Multiple R-squared:  0.7018, Adjusted R-squared: 0.6379
F-statistic: 10.98 on 3 and 14 DF,  p-value: 0.0005662
```

§2.9. PO example: comparison of models with/without centralization

- The fitted values $X_c\hat{\beta}_c$ (centered) and $X\hat{\beta}$ (un-centered) are the same, since they are the projection of $\boldsymbol{y}$ onto the same space. The residual standard error, R^2, and F-statistic are all the same.
- The estimates and their standard deviations are different:

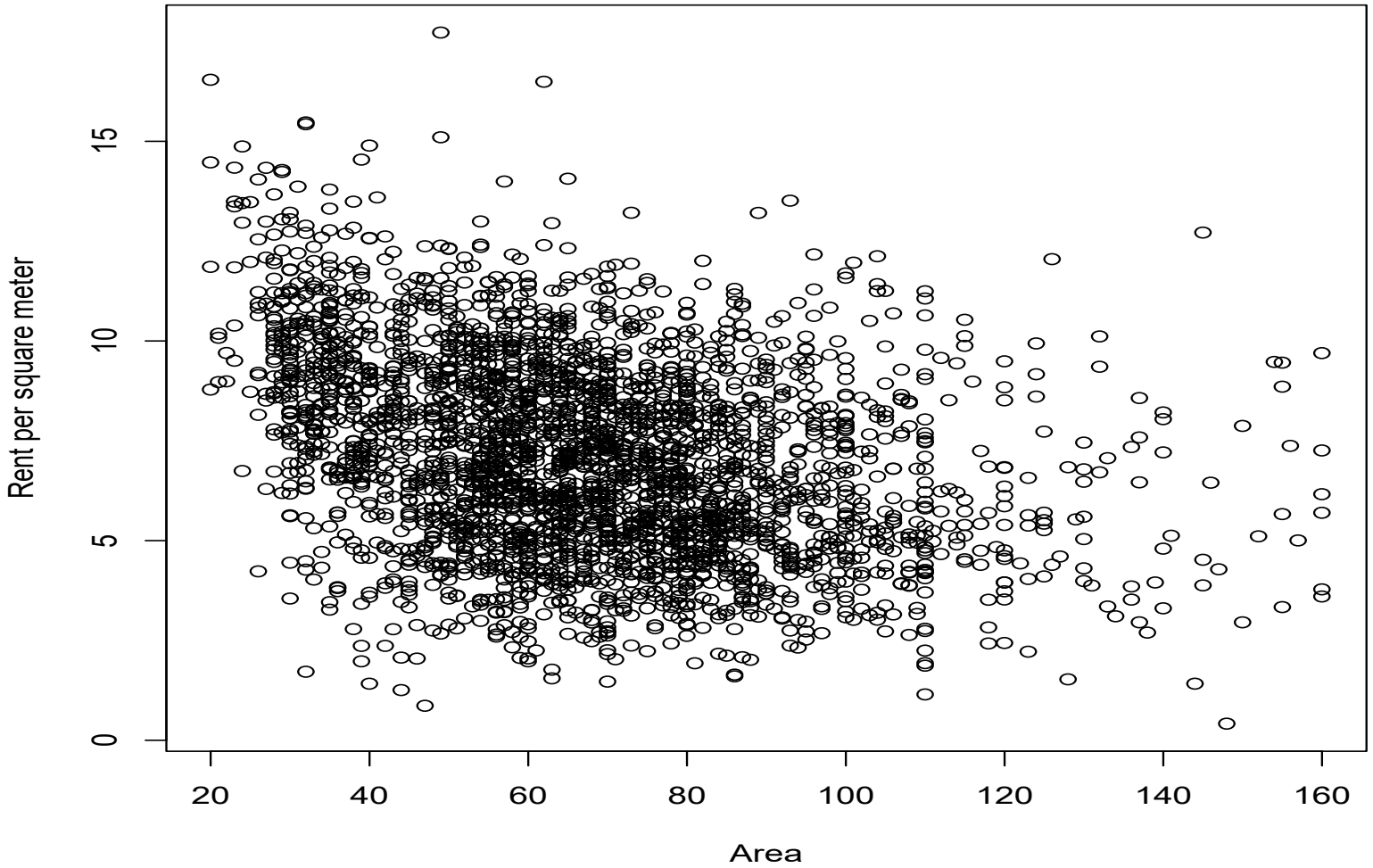
```
Centralized:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.047e+02  1.168e+00  89.664  < 2e-16
I(x.c)      -4.592e-01  1.562e-01  -2.939 0.010776
I(x.c^2)    -2.771e-03  5.265e-03  -0.526 0.606993
I(x.c^3)     2.246e-03  4.895e-04   4.588 0.000422
Non-centralized:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.511e+01  2.676e+01  -0.938 0.363924
x            1.054e+01  2.246e+00   4.694 0.000345
I(x^2)      -2.723e-01  5.897e-02  -4.617 0.000399
I(x^3)       2.246e-03  4.895e-04   4.588 0.000422
```

- The fitting without centralization incurs much larger variances than the centralized one. This is because there is severer multicollinearity in the non-centralization case.
- Question: why the coefficient of the cubic term has the same estimation in both cases?

§2.10. Munich rent index (RI): inverse transformation

- The X-terms in a LRM can be any proper transformation of physical variables. The Munich rent index data described below is used for an illustration.
- The Munich rent index data contains nine variables including `rentsqm`: rent per square meters, `area`: area of the rented house. The data is provided in the excel file `rent99.csv` on Canvas.
- We consider how the variable `area` (x) can be used for predicting `rentsqm` (y).
- The scatter plot helps to determine the transformation of x to be used in the regression model.

§2.11. RI example: scatter plot of the data



§2.12. Models suggested by the scatter plot

- The scatter plot shows a downward linear trend but with a slight curvature.
- Two models can be investigated: the simple LRM,

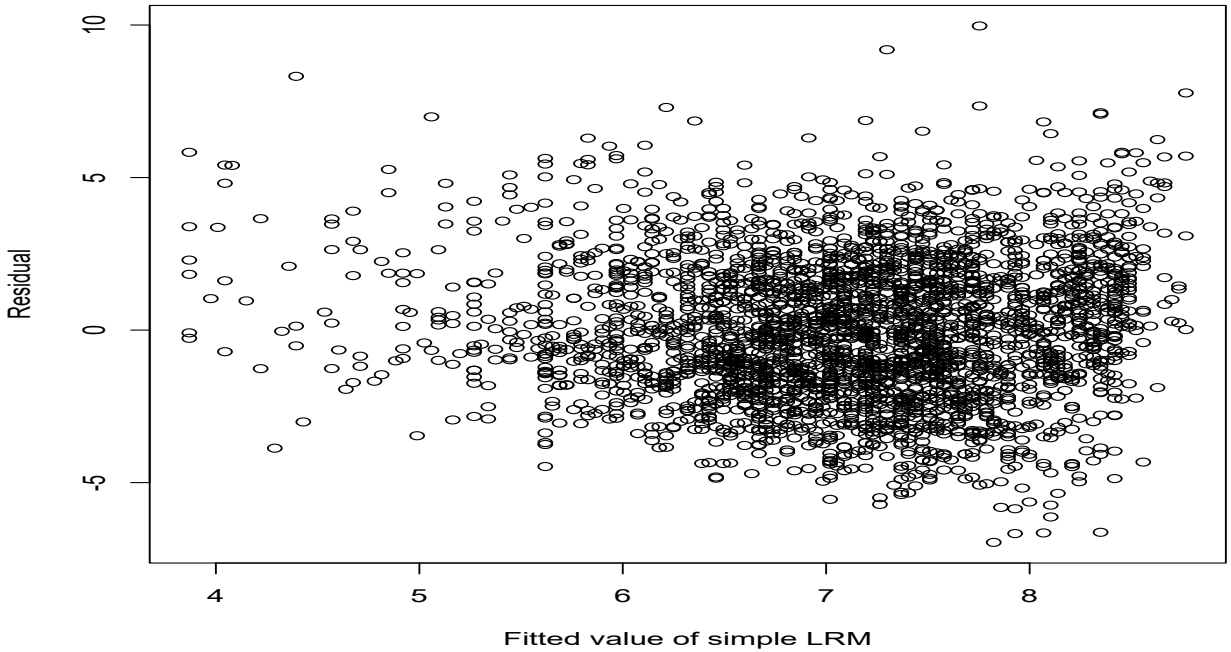
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

and the inverse predictor model,

$$y_i = \beta_0 + \beta_1 (1/x_i) + \epsilon_i.$$

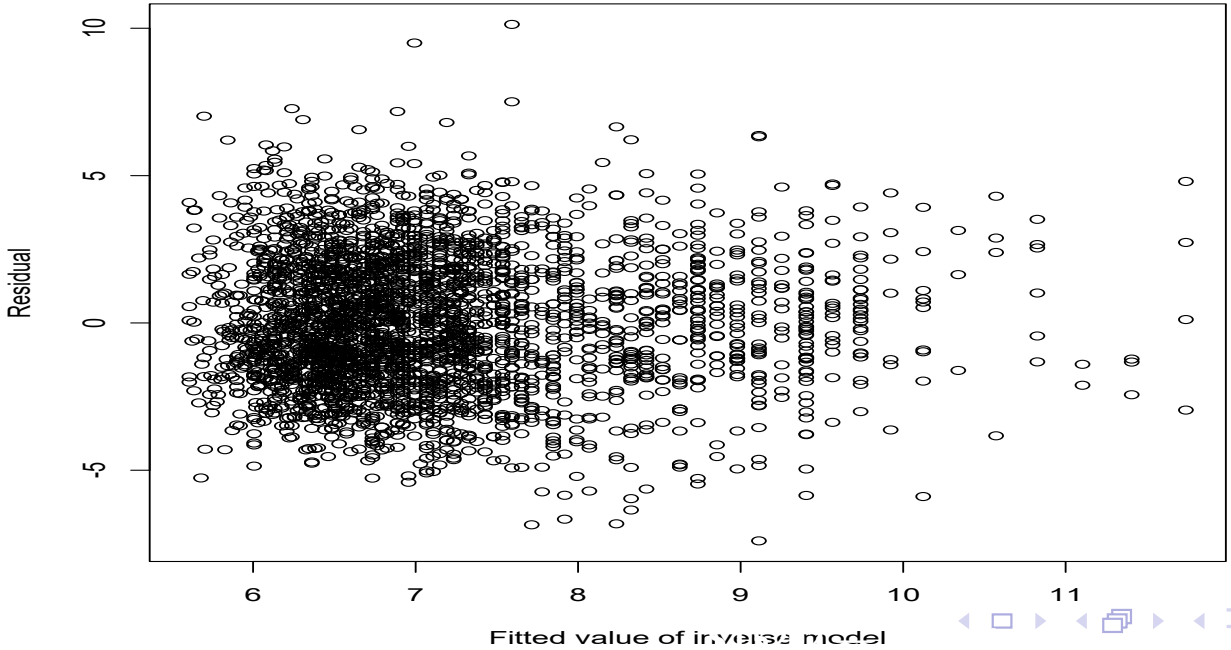
§2.13. RI example: simple LRM fitting

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.46883    0.12426   76.20   <2e-16
area        -0.03499    0.00174  -20.11   <2e-16
Residual standard error: 2.291 on 3080 degrees of freedom
Multiple R-squared:  0.1161,  Adjusted R-squared:  0.1158
```



§2.14. RI example: inverse model fitting

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.7321     0.1084   43.65   <2e-16
I(1/x)      140.1783     5.9287   23.64   <2e-16
Residual standard error: 2.241 on 3080 degrees of freedom
Multiple R-squared:  0.1536,  Adjusted R-squared:  0.1533
F-statistic:   559 on 1 and 3080 DF,  p-value: < 2.2e-16
```



§2.15. RI example: Comparison of models

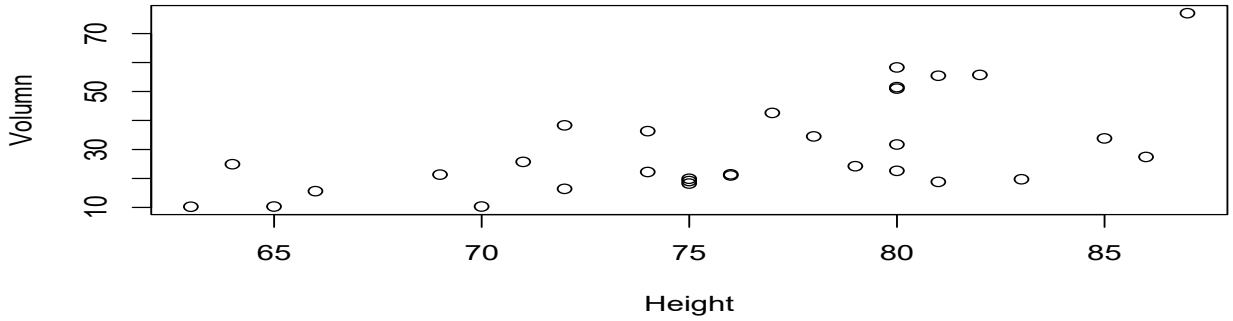
- The simple linear model does not catch the curvature feature exhibiting in the data. The curvature still remains in the residual plot.
- The inverse model well captures the curvature. No any trend and curvature is shown in the residual plot.
- The simple linear model explains about 12% of the total variation in Y, the inverse model explains about 15% of the total variation.
- The inverse model is more appropriate than the simple linear model.

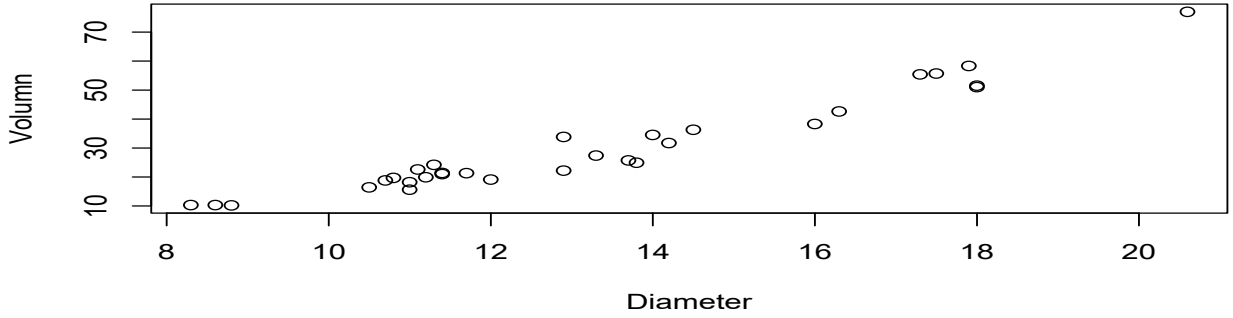
§2.16. Cherry tree (CT) example: log transformation

- The data set is originally provided in the statistics software `Minitab`. The data consists of the volume (v), diameter (d) (at $4'6''$ above ground level) and height (h) of 31 black cherry trees. The interest is to derive a formula to predict tree volume from diameter and height.
- The data is reproduced and provided in the file `trees.txt` on Canvas.
- The volumn formula $v = h\pi(d/2)^2$ suggests the following log transformation model:

$$log(v) = \beta_0 + \beta_1 \log h + \beta_2 \log d + \epsilon.$$

§2.17. CT example: scatter plots of the data

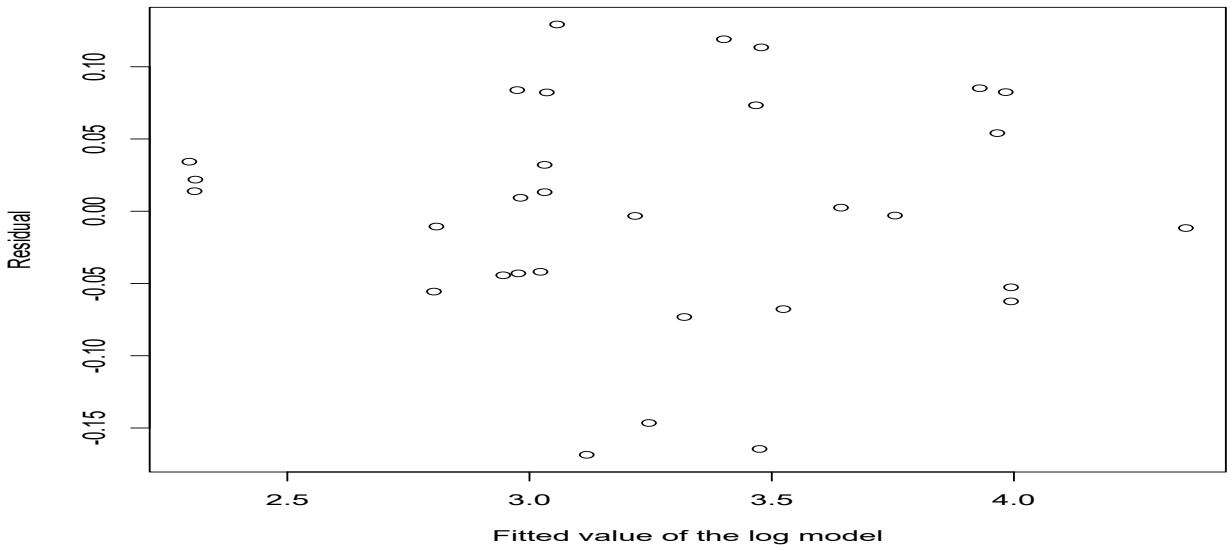




In both scatter plots, the trend is not exactly linear, both shows a curvature. It also suggests the log transformation model.

§2.18. CT example: fitted results

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -6.63162    0.79979  -8.292 5.06e-09
log(h)       1.11712    0.20444   5.464 7.81e-06
log(d)       1.98265    0.07501  26.432  < 2e-16
Residual standard error: 0.08139 on 28 degrees of freedom
Multiple R-squared:  0.9777,  Adjusted R-squared:  0.9761
F-statistic: 613.2 on 2 and 28 DF,  p-value: < 2.2e-16
```



§2.19. CT example: conclusions

- No pattern and trend show up in the residual plot of the log transformation model. The log transformation model seems a good fit.
- The log transformation model explains about 98% of the total variation in $log(v)$.
- Both the height and diameter have significant effect on the volumn of the black cherry trees.

§2.20. Piece-wise linear models

- In certain problems, the relation between the response Y and physical variable X might be different over different ranges of X, though the relationship is still locally linear.
- Introduce the auxiliary X truncated at a point $X = c$:

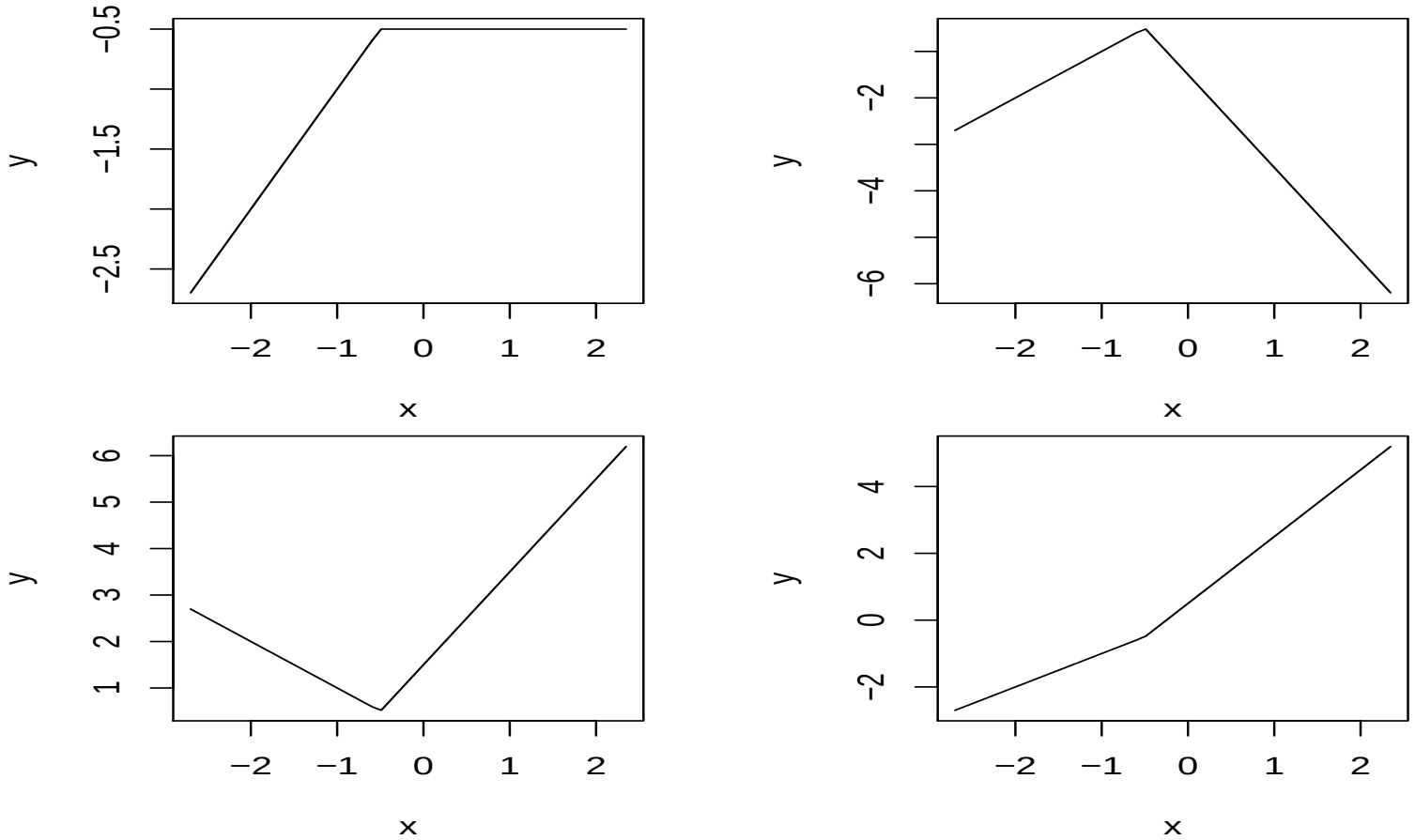
$$\tilde{X} = (X - c)^+ = \begin{cases} X - c, & \text{if } X - c \geq 0 \\ 0, & \text{otherwise .} \end{cases}$$

- Augmenting the original model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ by the auxiliary X, the model becomes:

$$Y_i = \beta_0 + \beta_1 X_i + \alpha_1 \tilde{X}_i + \epsilon_i.$$

- The resulted regression function is piece-wise linear in X. It changes the slope of the linear lines at the point $X = c$ from β_1 to $\beta_1 + \alpha_1$.

§2.21. Demonstration of piece-wise linear functions



§2.22. More general piece-wise linear models

- More pieces can be considered by introducing more auxiliary X's truncated at different points, say, $c_1 < c_2$ and so on. The model becomes

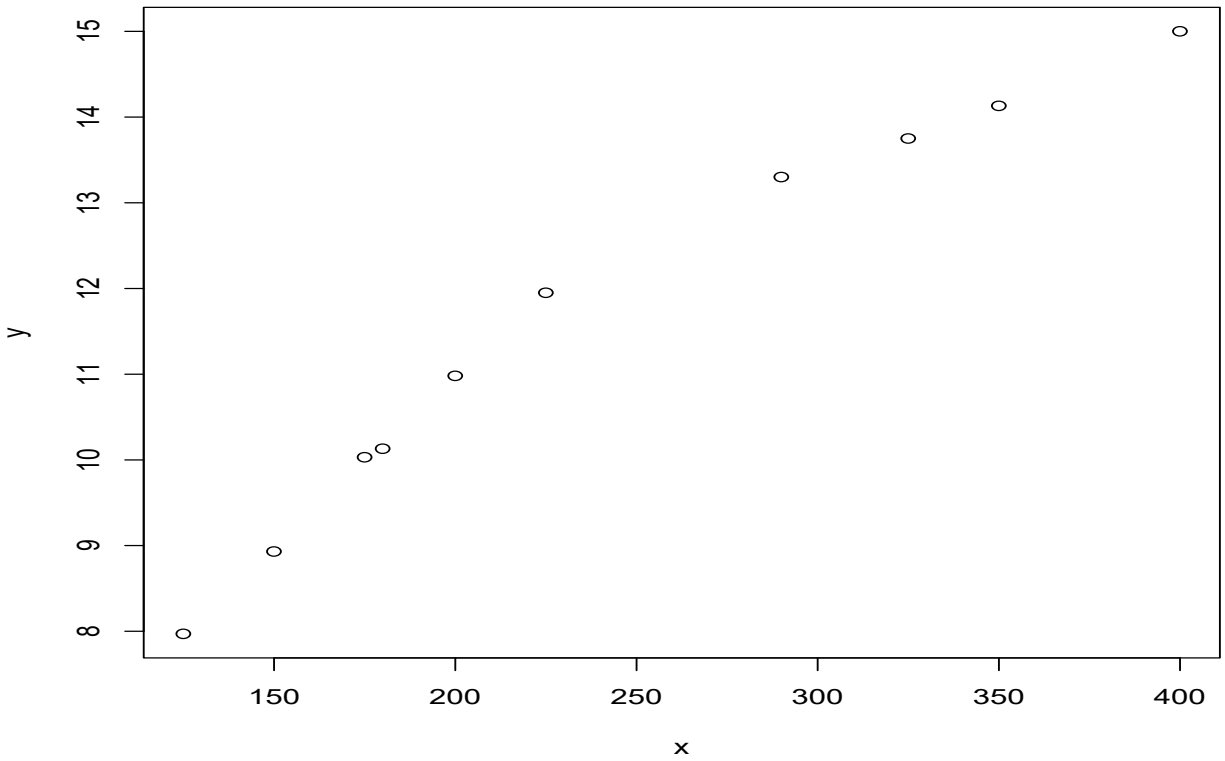
$$Y_i = \beta_0 + \beta_1 X_i + \alpha_1 \tilde{X}_{1i} + \alpha_2 \tilde{X}_{2i} + \epsilon_i.$$

  The regression line changes slope at $X = c_1$ and $X = c_2$ in turn from β_1 to $\beta_1 + \alpha_1$ to $\beta_1 + \alpha_1 + \alpha_2$.
- The same idea can be applied to the case of multiple physical variables.
- The piece-wise linear models enhance the ability of LRM to model more practical situations. It can also be used for diagnosis of the original model.
- Whether of not the regression line change its slope at a specified point, say c_2, can be tested by testing $H_0 : \alpha_2 = 0$.

§2.23. Shipment Example

- An electronics company periodically imports shipments of a certain large part used as a component in several of its products.
- The size of the shipment varies depending upon production schedules. For handling and distribution to assemby plants, shipments of size 250 thousand parts or less are sent to warehouse A; larger shipments are sent to warehouse B since this warehouse has specialized equipment that provides greater economies of scale for large shipments.
- The data set contains information on the cost (y) of handling the shipment in the warehouse (in thousand dollars) and the size of the shipment (x, in thousand parts). The data is provided in `shipment.txt` on Canvas. It is to be found how y is related to x.

§2.24. Scatter plot of the data



There is an obvious change of slope around 250. A piecewise linear regression model with a change point at $x = 250$ could be appropriate. This also complies with the background information.

§2.25. R codes and fitted results

```
dat =read.table("D://Rsession/shipment.txt",header=TRUE)
y=dat$y; x=dat$x1; z=(x-250)*(x>=250)
fit = lm(y~x+z); summary(fit)
```

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.2139257  0.1785735   18.00 4.04e-07 ***
x            0.0384599  0.0009554   40.25 1.52e-09 ***
z           -0.0247734  0.0016460  -15.05 1.37e-06 ***
Residual standard error: 0.09303 on 7 degrees of freedom
Multiple R-squared: 0.9988, Adjusted R-squared: 0.9985
F-statistic:  2938 on 2 and 7 DF,  p-value: 5.813e-11
```

The coefficient of z, the auxiliary variable, has a p-value 1.37e−06 and is significantly nonzero. Hence the change of slope is significant.